2 Bundles

In physics, the concept of fields is of prime importance. These are not to be confused with the fields mathematicians talk about. Mathematician's field is something like the real numbers or complex numbers. When physicists talk about fields, we have something like an electric field in mind. A naive mental picture is a bunch of arrows, one at each point in a space. How do we make this idea precise? One could start from making clear statements about where these arrows are coming from, i.e, the arrows are from a vector space. But assigning a vector from some random vector space to a point on the manifold will be artificial. What would be natural is to attach at each point $p \in M$, a vector from say the tangent space T_pM. One immediate issue with this idea is that then taking derivatives has to be defined carefully. Because, for a derivative, we need to compare vectors from two different points, and the vector spaces at different points are completely different. The theory of bundles is what makes all of this notions rigorous.

2.1 What is a Bundle?

Take a circle. To each point on the circle, attach a real line. What you have got in hand is an example of a line bundle! Let us make this precise. The circle is the manifold we used as the base to attach real lines. So to make a bundle, we need a manifold that we will call the base space, M. Now, we need the real lines. We will call them the fibers. The thing we get when we attach fibers at each point on the base space, we will call it the total space, E. One thing that we necessarily should have is, if we are at some fiber F on the total space E, then there should be a way to tell where this fiber is attached to the base space. This, we will call the projection map $\pi : E \to M$.

This is the general idea. Mathematicians like to take the general idea, and cast it in the most economic form. They came up with the following as the definition of a bundle :

> A bundle consists of a base space M, total space E and a projection map $\pi : E \to M$

In the example of the circle and real line, the base space is the circle, the total space is $S^1 \times \mathbb{R}$. Any general point on the total space will look like (θ, x). Then the projection map, $\pi : (\theta, x) \mapsto \theta$.

Where is the fiber in this definition? fiber is all the points in the total space that projects to the same point in the base space. That is the fiber at $p \in M$, denoted by E_p is defined as,

$$E_p := \left\{ q \in E \,\middle|\, \pi(q) = p \right\} \tag{2.1}$$

It is clear that if we take all the fibers together (bundle the fibers together!) we get the total space back.

2.2 The Tangent Bundle

Let us now get back to the vector field. For this, we will first make a bundle, called the tangent bundle. This is easy, we just have to choose the fiber at p to be the tangent space T_pM. Then the total space of the bundle is,

$$TM = \bigcup_{p \in M} T_pM \tag{2.2}$$

TM is the collection of all tangent vectors from all points in M. The projection map will take a vector and find its tangent space.

Now, we have to show that the TM is actually a manifold. Only then would it be a bundle. If the manifold M is n dimensional, it turns out that TM is a $2n$ dimensional manifold. It is fairly easy to understand. If we think of tangent vectors as velocities, then TM consists of velocities and the associated base space point. Velocity has n components, and the base space point is also an n component object. Therefore, it is easy to intuit that a point is base space looks like,

$$P \in TM : P \sim (v^1, v^2, \cdots, v^n, x^1, x^2, \cdots, x^n) \tag{2.3}$$

We can make this precise. Let us take charts in M, $\phi_\alpha : U_\alpha \to \mathbb{R}^n$. We want to define charts on TM, that is constructed out of the charts in M. The connection between TM and M is the projection map. Therefore, it is natural to think of a subset of TM, which under projection falls completely in the open set U_α. i.e,

$$V_\alpha = \left\{ v \in TM \,\middle|\, \pi(v) \in U_\alpha \right\} \tag{2.4}$$

Now, we need to show that these are open sets. For that we have to show a multitude of things, but lets just think about one of them, the intersection rule. The intersection of two V_α's should be also another V_α. This is easy to show, using U_α's openness. Overall, this is an easy exercise.

Now, we need maps from V_α to $\mathbb{R}^{2n}$. Half of it is easily done. For a $v \in V_\alpha$, find where this v is attached on M, and map that point to $\mathbb{R}^n$. The other half, as we intuited before, are the velocities. Since these velocity vectors in V_α is attached to some point in U_α, we can push forward the components to $\mathbb{R}^n$ and have the full map! So we define the map,

$$\psi_\alpha : v \in TM \longmapsto (\phi_\alpha(\pi(v)), (\phi_\alpha)_\star v) \in \mathbb{R}^n \times \mathbb{R}^n \tag{2.5}$$

It is clear that ψ_α is a smooth map, since ϕ_α and $\phi_{\alpha\star}$ are both smooth. Likewise, ψ_α maps open sets to open sets. Therefore, (V_α, ψ_α) are charts, and hence TM is indeed a manifold!

2.3 Trivial vs Locally Trivial

It turns out most of the bundles that we are interested in physics are something called locally trivial. Let us understand what this means. First, what is a trivial bundle? A trivial bundle is something that we can write as *base space* $\times$ *fiber*. The first example of $S^1 \times \mathbb{R}$ was one such. To generalise this, we can just think of base space M and something called a standard fiber F and construct the total space as,

$$E = M \times F \tag{2.6}$$

Clearly, the tangent bundle is an example of a non-trivial bundle. The fibers at different points are obviously different. However, the tangent planes of close by points are almost "identical/parallel". This is where the idea of locally trivial bundle comes in. A bundle is locally trivial if the bundle is trivial on the neighborhood of a point in the base space. For a mathematician, this is too vague. Sure, we can think of the local neighborhood in a bundle. This is $E\Big|_U = \Big\{q \in E \Big| \pi(q) \in U\Big\}$ for $U \subset M$. This is called a restriction of a bundle. If U is a local neighborhood in M, how can we say that this restriction "looks the same" as a trivial bundle? For this, we need the concept of isomorphism of bundles.

Let us take two bundles, $\pi : E \to M$ and $\pi' : E' \to M'$. We say that there is a morphism between these bundles if there are maps $\psi : E \to E'$ and $\phi : M \to M'$, and these maps are consistent. By consistent what we mean is that ψ maps fiber $E_p \in E$ to $E'_{\phi(p)} \in M'$. In more words, if we take a point $p \in M$ and map it to M' and find the corresponding bundle, and alternatively take the fiber at p on E and map it to E', we should get the same fiber. Now, if these two maps are diffeomorphisms, then we call the morphism an isomorphism.

A quick example can be given for tangent bundles. Consider M and M' with a map ϕ between them. Then, we can just choose $\psi = \phi_\star$, the push forward, and we get a bundle isomorphism.

Now we can define locally trivial bundles precisely. A bundle $\pi : E \to M$ is locally trivial with standard fiber F, if for all $p \in M$, there is a neighborhood U, such that there is a bundle isomorphism,

$$\phi : E\Big|_U \to U \times F \tag{2.7}$$

This isomorphism ϕ will be called local trivialisation.

An example is in order. We can show that the tangent bundle is locally trivial. This is easy. For any neighborhood $U_\alpha \subset M$, we have charts $V_\alpha : TM \to \mathbb{R}^n \times \mathbb{R}^n$. These are the restrictions. This immediately gives us the local trivialisation. It is,

$$\phi(v \in V_\alpha) = (\pi(v), \phi_{\alpha\star}v) \in U \times \mathbb{R}^n \tag{2.8}$$

The standard fiber here is $\mathbb{R}^n$.

The Möbius strip is another example of a locally trivial bundle. Locally, it looks like a trivial bundle with standard fiber $\mathbb{R}$. But it is not just $S^1 \times \mathbb{R}$, it has a twist in it.

2.4 Vector Bundles

These are the most important kind of bundles for physicists. The definition is what we would expect,

> A vector bundle is a locally trivial bundle $\pi : E \to M$ with the standard fiber being an n dimensional vector space.

Also, we would require that the local trivialisation would respect the vector space structure, which is linearity. Therefore, we will amend the definition as,

> A vector bundle is a locally trivial bundle $\pi : E \to M$ with the standard fiber being an n dimensional vector space, with $\forall p \in M : \exists U$ such that the local trivialisation $\phi : E|_U \to U \times \mathbb{R}^n$ maps the fibers E_p to $\{p\} \times \mathbb{R}^n$ linearly.

The first example is again the tangent bundle. Tangent bundle is locally trivial, and each fiber T_pM is an n dimensional vector space. The local trivialisation would take a vector $v_p \in E_p = T_pM$ and sends it to $\{p\} \times v^\mu$, where v^μ are the components of the vector. It is clear that this is a linear mapping.

The next example is something called a line bundle. We started this to intuit what a bundle is. A line bundle is a vector bundle with standard fibers being mapped to the real line. For example, the Möbius strip is a line bundle.

There is one subtlety to keep in mind. If we have a vector bundle, then there would be several local trivializations possible around each point. Any point p will in general be part of multiple open sets in M, and therefore we have multiple local trivializations. But we will make sure that the laws of physics will be independent of this choice of isomorphism. This would be called the principle of gauge invariance.

Also, it goes without saying that we can replace $\mathbb{R}$ with $\mathbb{C}$ to get complex vector bundles. But this introduces no new complications, so we will stick with $\mathbb{R}$.

2.5 Fields : The Section of a Bundle

The fields we use in physics are actually sections of a bundle. The way to picture this is to go back to our line bundle on a circle. To get a section of this bundle $S^1 \times \mathbb{R}$, we just have to choose one value in $\mathbb{R}$ for each value of $\theta \in S^1$. Let us make this precise.

$$\text{The section } s : M \to E \text{ of a bundle } E \xrightarrow{\pi} M \text{ is such that } \forall p \in M : s(p) \in E_p$$

One immediate property of sections is that if we project the section at $p \in M$, we get back p. That is, $\pi \circ s = \mathrm{id}_M$.

In the case of a trivial bundle $E = M \times F$, the section would look like $s(p \in M) = (p, v)$ where $v \in F$. This is just a roundabout way to define a function $f : M \to F$. The function corresponding to the section is $f(p) = v$. The section $s(p) = (p, f(p))$ is then just the graph of the function. The benefit of generalising a function to a section is that the idea of section would work even for non trivial bundles!

For example, if we make a section of the tangent bundle, we would have,

$$\text{for } p \in M : s(p) = (p, v_p), \quad \text{where} \quad v_p \in T_pM \tag{2.9}$$

Therefore, the vector field that we love is just a section of the tangent bundle!

We can define addition and $C^\infty(M)$ multiplication on sections of vector bundles very naturally. Let s and s' be two sections of the vector bundle $E \xrightarrow{\pi} M$. Then,

$$(s + s')(p) = s(p) + s'(p) = (p, v_p + v'_p) \tag{2.10}$$

and,

$$f \in C^\infty(M) : (fs)(p) = f(p)s(p) = (p, f(p)v_p) \tag{2.11}$$

We will denote the set of all sections of the vector bundle $E \xrightarrow{\pi} M$ to be $\Gamma(E)$. From the above definitions, it is clear that $\Gamma(E)$ is a module[1] over $C^\infty(M)$.

Since the set of sections of a vector bundle form a $C^\infty(M)-$module, it is natural to think to define a basis on $\Gamma(E)$. The definition is no different that we know. The sections $e_1, e_2, \cdots, e_n \in \Gamma(E)$ will be called a basis if any section $s \in \Gamma(E)$ can be written as,

$$s = s^i e_i, \quad \text{where} \quad s^i \in C^\infty(M) \tag{2.12}$$

But, this is too quick! That is because of the following theorem:

Theorem : A vector bundle has a basis of sections if and only if it is a trivial bundle.

Proof : First, assume that there is a basis, $\{e_i\}$. Then, we can define a function from the trivial bundle $M \times \mathbb{R}^n$ to E as,

$$\psi(p, v) := v^i e_i(p) \in E \quad \text{where} \quad p \in M, v \in \mathbb{R}^n \tag{2.13}$$

This is a bundle isomorphism. Therefore, the bundle $E \xrightarrow{\pi} M$ is trivial and can be written as $M \times \mathbb{R}^n$.
Now assume the converse, that the bundle is trivial. Then there is an isomorphism $\psi : M \times \mathbb{R}^n \to E$. On $M \times \mathbb{R}^n$, we have the natural basis of sections, which is $e_i = \Big(p, \big(\frac{\partial}{\partial x^i}\big)_p\Big)$. We just have to "push forward" this basis to E.

Let us understand this with the help of an example. Take a non trivial line bundle, say the Möbius strip. The base space here is $M = S^1$. We first have to define the Möbius strip precisely. The easiest way is to us a strip of paper, and glue on of the ends to the opposite

[1] A module is generalisation of vector space. A vector space will have an underlying field, where here field means the mathematicians field. For a module, the underlying structure is not a field, but a ring. A ring shares almost all properties with the field, except there are not multiplicative inverse.

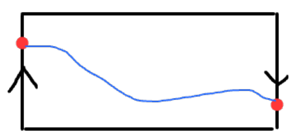
end, but with a twist. To be more precise, take a strip $[0, 2\pi] \times [-1, 1]$. Let us call the coordinates in this strip (x, y). Now, make the identification $(2\pi, y) \sim (0, -y)$. But if we want to think of the Möbius strip as locally a trivial bundle with fiber $\mathbb{R}$, then we have to change $[-1, 1]$ to $\mathbb{R}$. Then, the identification also should be modified. We know if we go twice around the Möbius strip, we get back to the same point. Therefore, if we take the set $\mathbb{R} \times \mathbb{R}$, then the appropriate identification is $(x + 2\pi k, (-1)^k y) \sim (x, y)$ for $k \in \mathbb{Z}$. Now, let us take some section, $s : S^1 \to (\mathbb{R} \times \mathbb{R})/\sim$. The section in general will look like,

$$s(x) = (e^{ix}, y) \tag{2.14}$$

Where the 2π identification of x has been manifest. If we go around the circle once, we must have,

$$s(x + 2\pi) = (e^{i(x+2\pi)}, y) = (e^{ix}, -y) \tag{2.15}$$

The equality is due to the identification. But $s(x)$ and $s(x + 2\pi)$ are both the same, because x and $x + 2\pi$ are identified. Hence, any section on the strip has to start from (x, y) and end at $(x + 2\pi, -y)$ (if we think of the plane). A figure of the same is given below,



The red dot at opposite ends have y positions negatives of each other. Therefore, every section should pass through $y = 0$ at least once! That means all sections vanish at at least one point.

This shows that the Möbius strip do not have a basis. To see this, take a section s and write this in terms of the basis. Let us assume that none of the basis sections vanish at p, where the section s vanishes. Then at p, a linear combination of the basis is equal to zero without the coefficients not equal to zero. Hence, the basis is not linearly independent, which violates the assumption of having a basis. Hence, Möbius strip do not have a basis of sections, hence is non trivial!

We will be interested in locally trivial bundles. Therefore, we can locally define basis sections and work with it.

2.6 Construction of Vector Bundles

We will now describe a few ways to construct new vector bundles given a vector bundle. These are going to be all the kinds of vector bundles that we will need while doing gauge theory.

Dual Vector Bundle

Let $E \xrightarrow{\pi} M$ be a bundle. Then, the fibers are vector spaces E_p. Take the dual of these fibers, E_p^*. Then, construct the bundle total space E^* as the union of these dual spaces. The projection map gets a similar definition, takes a cotangent vector from E_p^* to the point $p \in M$. Since the vector bundle E is locally trivial, we can have a basis of sections locally. Then, by the properties of dual vector spaces, we know that there are canonical dual basis vectors in the dual space. We can use them as the basis for sections of the bundle E^*.

Given sections $s : M \to E$ and $\lambda : M \to E^*$, we naturally have a smooth function on M defined to be $\lambda \circ s$. At each point, the section s gives a vector, and the section λ gives a linear functional that takes the vector to a real number. So we can define this function as,

$$(\lambda \circ s)(p) := \lambda(p)(s(p)) \tag{2.16}$$

A quick example would be the cotangent bundle, denoted by T^*M. The sections of the cotangent bundle are clearly one forms.

Direct Sum Bundle

Given two bundles E and E', the direct sum bundle can be defined to be,

$$\pi : E \oplus E' \to M \tag{2.17}$$

with fiber at p to be $E_p \oplus E'_p$. We know that this would be a vector bundle because direct sum of manifolds is a manifold, and direct sum of vector space is a vector space. Given local trivializations $\phi : E|_U \to U \times \mathbb{R}^n$ and $\phi' : E'|_{U'} \to U' \times \mathbb{R}^m$, we can define a local trivialisation on the direct sum bundle to be,

$$\psi : (E \oplus E')|_{U \cap U'} \to \mathbb{R}^{m+n} \tag{2.18}$$

One might wonder how can we intersect U and U'. But remember, U and U' are open sets of M, which is the base space for both bundles.

Direct Product Bundle

Given two bundles, we can construct,

$$\pi : E \otimes E' \to M \tag{2.19}$$

with the fiber at p to be $E_p \otimes E'_p$. This is also a bundle because $E \otimes E'$ is a manifold, and is a vector bundle because $E_p \otimes E'_p$ is a vector space. The local trivialization can be defined in a similar way as previous case.

Exterior Algebra Bundle

Let $E \xrightarrow{\pi} M$ be a vector bundle. Then, for each fiber E_p, we can define the exterior algebra, ΛE_p. Define the set,

$$\Lambda E = \bigcup_{p \in M} \Lambda E_p \tag{2.20}$$

This set is a manifold. To show this, use charts in the total space E - $\psi_\alpha : U_\alpha \to \mathbb{R}^n$ where $U_\alpha \subset E$ are the open sets - to define a chart map on ΛE. Remember, the chart map ψ_α pushes forward the vectors on the fiber to the tangent vectors in real numbers. A general element in ΛE will look like $(p, \text{form}) = (p, v_1 \wedge v_2 \wedge \cdots)$. Then, we can define the chart map $\Psi_\alpha : \Lambda E \to \Lambda \mathbb{R}^n$ to be the wedge product of push forwarded vectors,

$$\Psi_\alpha(p, v_1 \wedge v_2 \wedge \cdots) := (\psi_\alpha(p), \psi_{\alpha\star} v_1 \wedge \psi_{\alpha\star} v_2 \wedge \cdots) \tag{2.21}$$

The same chart maps give the local trivialization.

We can write the exterior algebra bundles as the direct sum,

$$\Lambda E = \bigoplus_{i=0}^{n} \Lambda^i E \tag{2.22}$$

where $\Lambda^i E$ is the bundle of $i-$forms.

Also, given two sections in the exterior algebra bundle, we can define a wedge product of sections in a natural way,

$$\omega, \mu \in \Gamma(\Lambda E) : (\omega \wedge \mu)(p) := \omega(p) \wedge \mu(p) \tag{2.23}$$

Form Bundle

This is a special case of the exterior algebra bundle. We know that T^*M is the bundle of one forms. Then the form bundle is just ΛT^*M.

Group Bundle

Since we are interested in locally trivial bundles, it would be nice to have a procedure that will let us glue together locally trivial vector bundles to create a non trivial vector bundle. Consider a manifold M and open sets $\{U_\alpha\}$ that covers M. We need a vector spaces to construct the vector bundle, call it V. Then, we have a set of trivial vector bundles, $U_\alpha \times V$. Since the open sets cover M, at least some of them necessarily have a non zero overlap. We want to glue these trivial bundles where there is non zero overlap, and end up with possibly a non trivial vector bundle. First, collect all trivial bundles together with a disjoint union,

$$\dot{\bigcup}_\alpha U_\alpha \times V \tag{2.24}$$

Then, we have to do the gluing and add it to the above set. Let $p \in U_\alpha \cap U_\beta$. Then, we have elements of the bundle to be $(p, v) \in U_\alpha \times V$ and $(p, v') \in U_\beta \times V$. We need to identify these elements using some way. Here is where the "G" in the title comes in. We will consider a group G and choose a function $g_{\alpha\beta} : U_\alpha \cap U_\beta \to G$, called a transition function. We will make use of a choice of representation of the group on V,

$$\rho : G \to GL(V) \tag{2.25}$$

and identify the elements v and v', if,

$$v = \rho(g_{\alpha\beta}(p))v' \tag{2.26}$$

We see that the gluing depends on a number of choices. The group G, the transition functions, and the representation.

The transition functions are actually not completely arbitrary. They have to satisfy certain conditions. The first of this is the obvious one. We should identify $(p, v) \in U_\alpha \times V$ with itself. That is, we have to identify v and $\rho(g_{\alpha\alpha}(p))v$. But we don't want to identify vectors from the same trivial vector bundles. Therefore, we must impose the condition,

$$\forall p \in U_\alpha : \rho(g_{\alpha\alpha}(p)) = 1 \tag{2.27}$$

The second condition is more involved. Consider $p \in U_\alpha \cap U_\beta \cap U_\gamma$. We must identify $(p, v) \in U_\alpha \times V$ with $(p, g_{\gamma\alpha}v) \in U_\gamma \times V$, where $\rho(g_{\gamma\alpha}(p))$ has been abbreviated to $g_{\gamma\alpha}$. Then $(p, g_{\gamma\alpha}v) \in U_\gamma \times V$ has to be identified with $(p, g_{\beta\gamma}g_{\gamma\alpha}v) \in U_\beta \times V$, which in turn should be identified with $(p, g_{\alpha\beta}g_{\beta\gamma}g_{\gamma\alpha}v) \in U_\alpha \times V$. Again, we do not want to identify two elements in the same vector bundle. Therefore, we must enforce what is called the cocycle condition,

$$\forall p \in U_\alpha \cap U_\beta \cap U_\gamma : g_{\alpha\beta}g_{\beta\gamma}g_{\gamma\alpha} = 1 \tag{2.28}$$

We can think of many more such conditions, but all of them will be derived from the above two.

An example of such a new condition is this : we have to identify $(p, v) \in U_\alpha \times V$ with $(p, g_{\beta\alpha} v) \in U_\beta \times V$ which in turn is identified with $(p, g_{\alpha\beta} g_{\beta\alpha} v) \in U_\alpha \times V$. In other words,

$$g_{\beta\alpha} = g_{\alpha\beta}^{-1} \tag{2.29}$$

Once this identification is done, we can include the equivalence class to the disjoint union to make E. Let the equivalence class be denoted by $[p, v]_\alpha$, for $(p, v) \in U_\alpha \times V$. By the identification, we have $[p, v]_\alpha = [p, g_{\beta\alpha} v]_\beta$ by definition. We can define the projection map uniquely, $\pi([p, v]_\alpha) = p$. Thus we have constructed the vector bundle! This is the G - Bundle, and the group G is called the gauge group.

The local trivializations are obvious, since this is what we started with. We have,

$$\phi_\alpha : E|_{U_\alpha} \to U_\alpha \times V \tag{2.30}$$

with,

$$\phi_\alpha([p, v]_\alpha) := (p, v) \tag{2.31}$$

One might wonder if this identification is unique, because the elements of the fiber are essentially equivalence classes. But the cocycle conditions saves the day,

$$(p, v) = \phi_\alpha([p, v]_\alpha) = \phi_\alpha([p, g_{\beta\alpha} v]_\beta) = \phi_\alpha \circ \phi_\beta^{-1}(\phi_\beta([p, g_{\beta\alpha} v]_\beta)) \tag{2.32}$$

we should then have,

$$\phi_\alpha \circ \phi_\beta^{-1} = \rho(g_{\alpha\beta}) \tag{2.33}$$

then using the cocycle condition, we see that,

$$(p, v) = \phi_\alpha([p, v]_\alpha) = \phi_\alpha \circ \phi_\beta^{-1}(\phi_\beta([p, g_{\beta\alpha} v]_\beta)) = (p, g_{\alpha\beta} g_{\beta\alpha} v) = (p, v) \tag{2.34}$$

Now, consider linear transformations on a fiber, $T : E_p \to E_p$. E_p "looks like" the vector space V in every local neighborhood. And the linear transformations of interest on V are the group representations $\rho(g)$. We will say that "T lives in G" if,

$$\exists g \in G : T([p, v]_\alpha) = [p, \rho(g) v]_\alpha \tag{2.35}$$

This is well defined, and independent of α. To see this, recall that $[p, v]_\alpha = [p, \rho(g_{\beta\alpha}) v]_\beta$ for $p \in U_\alpha \cap U_\beta$. Similarly,

$$[p, \rho(g) v]_\alpha = [p, \rho(g_{\beta\alpha}) \rho(g) v]_\beta \tag{2.36}$$

Therefore,

$$T([p,\rho(g_{\beta\alpha})v]_\beta) = T([p,v]_\alpha) = [p,\rho(g)v]_\alpha = [p,\rho(g_{\beta\alpha})\rho(g)v]_\beta \tag{2.37}$$

Change variables to $v' = \rho(g_{\beta\alpha})v$ and take $g' = g_{\beta\alpha}gg_{\beta\alpha}^{-1}$, then the last equation reads,

$$T([p,v']_\beta) = [p,\rho(g')v']_\beta \tag{2.38}$$

which again defines "T lives on G". It is clear from the definition that it makes sense to only ask if T is of the form $\rho(g)$, and not ask which specific g are we talking about. This depends on the choice of coordinates aka local trivialization.

The same definition can be made for T living in the Lie algebra $\mathfrak{g}$. T lives in $\mathfrak{g}$ if,

$$\exists t \in \mathfrak{g} : T([p,v]_\alpha) := [p, d\rho(t)v]_\alpha \tag{2.39}$$

Let us consider an explicit example to understand this whole procedure of gluing. We take the Möbius strip. Take the circle S^1 as the base space, and take two open sets U_1 and U_2 that covers the circle. We then have local line bundles,

$$U_\alpha \times \mathbb{R} \tag{2.40}$$

We want to glue them, but with a twist. To introduce the twist, we use the gauge group $\mathbb{Z}_2$. This is the cyclic group with elements $\{0,1\}$, and the rule of group multiplication is addition modulo 2. We will define a representation,

$$\rho : \mathbb{Z}_2 \to \mathbb{R} \tag{2.41}$$

as,

$$\text{for } x \in \mathbb{R} : \rho(0)x = x \quad \text{and} \quad \rho(1)x = -x \tag{2.42}$$

Then, we choose the transition function to be,

$$g_{11} = g_{22} = 0 \in \mathbb{Z}_2 \quad \text{and} \quad g_{12} = g_{21} = 1 \in \mathbb{Z}_2 \tag{2.43}$$

Here, there is a slightly confusing notational artefact. The cocycle condition requires $g_{\alpha\alpha} = 1 \in G$. But for $\mathbb{Z}_2$, the identity element is denoted by 0, and we are actually good. We can check the other cocycle conditions, for example,

$$g_{12}g_{21} = (1+1)\%2 = 0 = 1_{\mathbb{Z}_2} \tag{2.44}$$

We are now ready to glue. For this, we identify $(p,x) \in U_1 \times \mathbb{R}$ with $(p,\rho(g_{21})x) = (p,-x) \in U_2 \times \mathbb{R}$. We see the twist appearing. If instead we chose the transition functions to be $g_{21} = g_{12} = 0$ as well, then we would just end up with the trivial bundle $S^1 \times \mathbb{R}$.